Lecture 5

Lecture 5 develops extremal chiral partition functions and the corrected Farey-tail expansion of protected indices.

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Conventions

We retain $\Lambda = -1/\ell^2$, $[G] = [\ell]$, and the spatial boundary circle of circumference $2\pi\ell$. Except in the explicitly supersymmetric examples, the theory is bosonic Einstein gravity with equal central charges

$$c_L = c_R = c = \frac{3\ell}{2G}, \qquad k_{\mathrm{CS}} = \frac{\ell}{4G} = \frac{c}{6}, \qquad k \equiv \frac{c}{24} = \frac{k_{\mathrm{CS}}}{4}. \tag{1}$$

Here k_{CS} is the fundamental-trace level used in lecture 2; k uses the vector-trace normalization of [1]. Their factor of four is a trace convention, not a quantum correction. Until §1.1, k is just a real parameter. The relation to ℓ/G is the classical Brown–Henneaux normalization. In the quantum characters below, c denotes the physical central charge; any local renormalization of the coupling and vacuum energy is absorbed into that parameter. This convention does not specify a universal shift between a bare Chern–Simons level and c. In lecture 6 we impose no integrality assumption on c. [11][9, §1, footnote 1]

Write $h, \bar{h}$ for the plane eigenvalues of $L_0, \bar{L}_0$, so the vacuum has $(h, \bar{h}) = (0, 0)$. The cylinder energies and angular momentum are

$$E_L = h - \frac{c}{24} = \frac{\ell M + J}{2}, \qquad E_R = \bar{h} - \frac{c}{24} = \frac{\ell M - J}{2}. \tag{2}$$

Thus $\ell M = h + \bar{h} - c/12$ and $J = h - \bar{h}$; this is the orientation convention of lecture 3. Some references reverse the sign of J, and Witten often denotes the shifted cylinder generator itself by L_0. We translate both conventions into (2). [1, §3.1][8, §1]

For a real Euclidean boundary torus,

$$w \sim w + 2\pi \sim w + 2\pi\tau, \quad \tau = x + iy, \quad y > 0, \quad q = e^{2\pi i\tau}, \quad \bar{q} = e^{-2\pi i\bar{\tau}}, \quad \bar{\tau} = \tau^*. \tag{3}$$

The trace is

$$Z(\tau, \bar{\tau}) = \mathrm{Tr}_{\mathcal{H}} q^{E_L} \bar{q}^{E_R} = \mathrm{Tr}_{\mathcal{H}} e^{-2\pi y \ell M + 2\pi i x J}. \tag{4}$$

Here $\widehat{\beta} = \beta_{\mathrm{phys}}/\ell = 2\pi y$ and $2\pi x$ is a real angular twist. A real Lorentzian angular potential is obtained by analytic continuation: $\beta_L = \widehat{\beta}(1 - \widehat{\Omega})$, $\beta_R = \widehat{\beta}(1 + \widehat{\Omega})$, with $\widehat{\Omega} = \ell\Omega$, $\tau = i\beta_L/(2\pi)$ and $\bar{\tau} = -i\beta_R/(2\pi)$. Then τ and $\bar{\tau}$ need not be complex conjugates. This continuation is already needed for ordinary rotating thermodynamics; it should be distinguished from enlarging the set of bulk configurations. [7, §2] [checked]

1 Chern–Simons quantization and extremal conformal field theories

1.1 Global gauge data and the quantization of the central charge

The thermodynamic calculation of lecture 4 gives the leading contribution of an AdS vacuum or a BTZ black hole to a boundary partition function. It does not yet tell us which values of ℓ/G admit a quantum theory, or how the vacuum and black-hole states fit into a single Hilbert space. Both questions are unusually accessible in three dimensions: the Einstein action can be rewritten as a Chern–Simons action, whose consistency on topologically nontrivial gauge configurations can restrict its coupling. Since that coupling fixes the Brown–Henneaux central charge, a restriction in the bulk becomes a restriction on the possible boundary CFTs. [1, §§1–2]

To apply this observation, recall how the coupling enters. The dreibein e converts local Lorentz indices into spacetime indices, and ω is the spin connection. Combining them into the two connections of lecture 2 gives

$$A = \omega + \frac{e}{\ell}, \qquad \bar{A} = \omega - \frac{e}{\ell}, \qquad I = \frac{k_{\mathrm{CS}}}{4\pi} \int \mathrm{Tr}_{\mathbf{2}} \left(A \wedge \mathrm{d}A + \frac{2}{3} A^3 \right) - (A \longrightarrow \bar{A}), \tag{5}$$

with boundary terms chosen for the desired boundary conditions. When e is invertible, the flatness equations for A and $\bar{A}$ are the Einstein equations together with vanishing torsion. The coefficient $k_{\mathrm{CS}} = \ell/(4G)$ is the level in the fundamental two-dimensional trace. For the global gauge group $\mathrm{SO}_0(2,1)$, where the subscript selects the identity component, it is convenient to use its three-dimensional vector trace. Since $\mathrm{Tr}_{\mathbf{3}} = 4\mathrm{Tr}_{\mathbf{2}}$ on this Lie algebra, the same action has level $k = k_{\mathrm{CS}}/4$ and central charge $c = 24k$. [10][1, §2.2] [checked]

The classical equations make sense for continuously variable k. The additional quantum requirement is that the phase e^{iI} be unambiguously defined on every allowed gauge configuration. This is not automatic for a Chern–Simons functional: the potential A can be defined only in patches, even though its curvature and the physical connection are well-defined. The obstruction, and the reason it quantizes k, can be seen without the non-Abelian algebra by first considering $U(1)$. [1, §2.1]

Let M be a closed three-manifold. If an Abelian potential is a single globally defined one-form, its action is $k(2\pi)^{-1} \int_M A \wedge \mathrm{d}A$. More general allowed fields carry quantized flux: $F/(2\pi)$ has integer integrals over closed two-surfaces, and no single potential covers all of M. Mathematically such a gauge field is a connection on a line bundle, which is the data needed to patch local potentials by $U(1)$ gauge transformations. To define the action in this case, choose an auxiliary

four-manifold X with boundary M, and extend both the line bundle and its connection from M into X. Then use the globally defined curvature of the extended connection:

$$I_X = \frac{k}{2\pi}\int_X F\wedge F, \qquad \partial X = M. \tag{6}$$

For a globally defined potential this reduces to the three-dimensional formula by Stokes' theorem, since $F\wedge F = \mathrm{d}(A\wedge \mathrm{d}A)$. The auxiliary interior is a device for defining the action; it must not become extra physical information in the quantum amplitude. [1, §2.1]

To test this independence, extend the same boundary gauge field into a second four-manifold X'. Gluing X to X' with reversed orientation along their common boundary produces a closed four-manifold $Y = X\cup_M (-X')$. The two extended connections also glue because they agree on M. Their action difference is therefore

$$I_X - I_{X'} = 2\pi k\int_Y \left(\frac{F}{2\pi}\right)^2 = 2\pi k N, \qquad N\in\mathbb{Z}. \tag{7}$$

The integer N measures the flux topology of the glued bundle; it is the integral of the square of its first Chern class. Requiring $e^{iI_X} = e^{iI_{X'}}$ for all such choices gives $k\in\mathbb{Z}$ in the bosonic theory defined without a choice of spin structure. [1, §2.1] [checked]

This Abelian calculation determines the relevant topological condition for $\mathrm{SO}_0(2,1)$ as well. Its noncompact directions can be continuously contracted, leaving the compact rotation subgroup $\mathrm{SO}(2)\simeq U(1)$; the nontrivial bundle topology is carried by that subgroup. Restricting the vector trace to it gives the Abelian normalization above, up to an overall sign that does not change the integrality condition. Applying the condition to both independent connections, with equal levels for Einstein gravity, gives

$$G_{\text{gauge}} = \mathrm{SO}_0(2,1)\times\mathrm{SO}_0(2,1), \qquad k_L = k_R = k\in\mathbb{Z}, \qquad c_L = c_R = 24k. \tag{8}$$

Thus this gauge-theory definition selects a discrete sequence of candidate central charges rather than a continuously variable ℓ/G. [1, §§2.1–2.3]

The allowed gauge configurations depend on more than the Lie algebra. Replacing the group by a covering group changes which fluxes and bundles are allowed. For an independent n-fold cover of one factor, the condition in the normalization above becomes $k\in n^{-2}\mathbb{Z}$; the universal cover removes that factor's level quantization. Diagonal covers impose correlated conditions on the two levels, and specifying a spin theory changes the quantization condition too. Consequently (8) is tied to the stated global gauge group. [1, §§2.1–2.3]

This also identifies the limit of the gravitational inference. The Einstein action on real smooth nondegenerate metrics has no auxiliary four-manifold ambiguity. The gauge-theory argument allows a larger configuration space, including connections whose reconstructed dreibein is degenerate. We can use its discrete couplings to organize candidate quantum theories, but cannot infer from it that every completion of metric gravity must have $c\in 24\mathbb{Z}$. To learn whether these candidates can contain the required vacuum and black-hole states, we now turn to their boundary partition functions. [1, §2.4]

1.2 Holomorphic factorization and the vacuum gap

Fixing the central charge still leaves a large freedom in the spectrum. The Chern–Simons action suggests a further simplifying assumption: A and $\bar{A}$ couple to the two chiralities, so one can ask

whether their quantum states can be chosen independently. If so, the Hilbert space and its trace take the form

$$\mathcal{H} = \mathcal{H}_L \otimes \mathcal{H}_R, \qquad Z(\tau, \bar{\tau}) = Z_L(\tau) Z_R(\bar{\tau}). \tag{9}$$

We will impose the stronger requirement that each chiral factor is separately modular invariant. In a left-right symmetric theory the answer is then $|Z_L(\tau)|^2$. This makes the spectral problem holomorphic: one can first determine a function of τ alone and then form its product with the right-moving answer. [1, §§2.3–2.4]

This factorization is a restriction on the quantum theory, beyond the existence of two Virasoro algebras. In a general CFT, left and right representations are paired according to a multiplicity matrix: $Z = \sum_{i,j} N_{ij} \chi_i \bar{\chi}_j$. Here χ_i is the trace over one chiral representation and N_{ij} counts how often that pair occurs. Such a pairing need not reduce to a single product. Likewise, the two local Chern–Simons actions may be separate while the allowed connections share a bulk topology or a reality condition. We retain factorization as an ansatz whose consequences can be tested. [1, §2.4][7, §4.2]

There is an immediate consistency check with the level quantization. Under $T : \tau \mapsto \tau + 1$, the vacuum term of a chiral trace, $q^{-c/24}$, acquires the phase $e^{-2\pi i c/24}$. Separate modular invariance requires this phase to be one, so $c = 24k$ with integer k. The nonchiral trace would only require cancellation of the left and right phases; it would not give this condition on each central charge. Thus the discrete levels selected above are precisely compatible with the separately invariant chiral ansatz. More general holomorphic characters can transform by phases or mix with other characters, but we are not imposing that broader possibility here. [1, §2.3] [checked]

The remaining input is the spectrum that can already be identified in gravity. The AdS vacuum is not an isolated state under the full asymptotic symmetry group. Transformations that act nontrivially at the boundary carry Brown–Henneaux charges and create descendants of the vacuum. The Virasoro algebra and vacuum conditions are

$$[L_m, L_n] = (m-n)L_{m+n} + \frac{c}{12} m(m^2-1)\delta_{m+n,0}, \qquad L_n|0\rangle = 0 \quad (n \geq -1). \tag{10}$$

The modes L_{-n} with $n \geq 2$ therefore generate boundary-graviton states; $L_{-1}|0\rangle$ vanishes because the vacuum is invariant under the global conformal group. For $c = 24k > 1$, an ordered basis uses occupation numbers $N_n \geq 0$ and has weight $h = \sum_{n \geq 2} n N_n$. Summing these states gives the vacuum character, which is the generating function for the vacuum and its descendants:

$$\chi_{\text{vac}}(\tau) = q^{-k} \prod_{n=2}^{\infty} (1 - q^n)^{-1} = q^{-k}(1 + q^2 + q^3 + 2q^4 + 2q^5 + 4q^6 + \cdots). \tag{11}$$

This sector is fixed before introducing any black-hole microstates. It cannot be the entire chiral theory, because its character is not modular invariant. [1, §3.1] [checked]

Pure gravity supplies a physical reason to ask for the additional states to start as high as possible. Below the black-hole threshold there are no local matter fields in the theory, whereas classical BTZ solutions have $E_L, E_R \geq 0$. We therefore look for a completion in which states below threshold are already accounted for by the vacuum module. To obtain a definite chiral ansatz, we also place the threshold $E_L = 0$ in that module: its classical chiral black-hole entropy vanishes, and the first new primary is taken to have $E_L = 1$, or $h = k + 1$. A primary is a state annihilated by the positive Virasoro modes, so it starts a new representation rather than being another excitation of the vacuum. [1, §3.1]

The resulting condition is

$$Z_k(\tau) - \chi_{\text{vac}}(\tau) = O(q). \tag{12}$$

It fixes every term from the vacuum pole through q^0 and is called the extremal condition. The treatment of the zero-entropy threshold is additional quantum input; classical BTZ thermodynamics alone does not require it. Its usefulness is that, together with holomorphy and modular invariance, it will determine a unique candidate torus trace. We can then ask separately whether a CFT realizes that trace. [1, §3.1]

In a factorized theory a left primary can be tensored with the right vacuum. Such states lie outside the classical BTZ region in one chirality. Therefore chiral extremality in (12) and the nonchiral requirement "no nonvacuum primary below the BTZ threshold in either chirality" are different spectral assumptions. The latter is studied in lecture 6, with its descendant-induced threshold shift. This distinction follows directly from (9) and (2). [checked]

1.3 The Klein invariant and the Monster theory

To implement the extremal condition, we need modular invariant functions with a prescribed pole at the low-temperature cusp. The Klein invariant provides a function with just one simple pole, so its powers will build the higher vacuum poles required at $c = 24k$. An explicit expression uses the Eisenstein series E_4 and E_6, holomorphic modular forms of weights four and six. Their normalized Fourier expansions are

$$E_4(\tau) = 1 + 240\sum_{n\geq 1}\sigma_3(n)q^n, \qquad E_6(\tau) = 1 - 504\sum_{n\geq 1}\sigma_5(n)q^n, \tag{13}$$

where $\sigma_r(n) = \sum_{d|n} d^r$. They determine the Klein invariant

$$\begin{aligned} j(\tau) &= 1728\frac{E_4(\tau)^3}{E_4(\tau)^3 - E_6(\tau)^2}, \qquad \mathcal{J}(\tau) \equiv j(\tau) - 744, \\ \mathcal{J}(\tau) &= q^{-1} + 196884q + 21493760q^2 + 864299970q^3 + \cdots . \end{aligned} \tag{14}$$

We use $\mathcal{J}$ to avoid confusing this function with angular momentum J. The reason to introduce j is that it converts our modular-invariance constraint into a polynomial problem. It is unchanged by a change of torus cycle basis and distinguishes inequivalent complex tori. After including the degeneration $q \to 0$, the space of such tori is a sphere with j as its coordinate.[1] The chiral trace has no singularities at a finite torus modulus and has the vacuum pole q^{-k} at the degeneration. It must consequently be a polynomial in j of degree k, or equivalently in $\mathcal{J} = j - 744$. [1, §3.1]

At $k = 1$, the unique function with principal part q^{-1} and vanishing constant term is $\mathcal{J}$. There is an actual CFT realizing it: the Frenkel–Lepowsky–Meurman moonshine module $V^\natural$, usually called the Monster CFT. Its automorphism group, preserving its operator products and stress tensor, is the Monster finite simple group. A construction starts with 24 chiral bosons based on the Leech lattice, an even unimodular rank-24 lattice without vectors of squared length two. The reflection $X^i \mapsto -X^i$ projects out its 24 weight-one currents; a consistent orbifold also includes

[1] A point τ in the upper half-plane specifies a complex torus. Identifying τ and $\gamma\tau$ removes different choices of its cycle basis; the resulting one-complex-dimensional parameter space is the modular curve. It has a missing degeneration at $\operatorname{Im}\tau \to \infty$, where the torus becomes infinitely long. Adding this point, called the cusp, compactifies the parameter space. For the full modular group the compactified space has genus zero and j has a single simple pole at that cusp, so it serves as a coordinate on a sphere. A function holomorphic away from the cusp and with a finite-order pole there is then a function of j holomorphic at every finite j. On the sphere such a meromorphic function is a polynomial; its degree is its pole order at infinity. This is the mathematical input used in the extremal construction. [1, §3.1]

the projected twisted sector. The resulting holomorphic theory has $c = 24$ and no weight-one states. [12][1, §3.1]

The torus partition function and its interpretation are

$$Z_{V^\natural}(\tau) = \mathrm{Tr}_{V^\natural} q^{L_0 - 1} = \mathcal{J}(\tau), \qquad Z_{V^\natural \otimes \overline{V^\natural}} = |\mathcal{J}(\tau)|^2. \tag{15}$$

The q^{-1} term is the unique vacuum; absence of q^0 means absence of $h = 1$ states. At $h = 2$, one of the 196884 states is $L_{-2}|0\rangle$, and the remaining 196883 are Virasoro primaries forming the smallest nontrivial Monster representation. The chiral theory has central charge 24; in the nonchiral product each chirality has 24, not 12. [12][1, §3.1]

This example establishes that the extremal torus data can be realized at $k = 1$. The proposed identification of the product theory with pure Einstein gravity is additional. In particular $c = 24$ supplies no parametrically large ratio ℓ/G, and the factorized theory has the mixed-chirality states just described. [1, §3.1] [checked]

1.4 Constructing the candidates at larger central charge

For $k > 1$, the vacuum pole has higher order and the vacuum descendants fix several additional low-energy coefficients. The polynomial property of the Klein invariant turns these conditions into a finite calculation. Write

$$Z_k(\tau) = \sum_{r=0}^{k} a_r \mathcal{J}(\tau)^r. \tag{16}$$

Since $\mathcal{J} = q^{-1} + O(q)$, fixing the coefficient of q^{-k} first determines $a_k = 1$. Fixing the coefficient of q^{-k+1} then determines a_{k-1}, and continuing down to q^0 fixes all $k+1$ coefficients. This triangular procedure proves uniqueness under the holomorphy and gap assumptions. It does not determine operator products. [1, §3.1] [checked]

For example, $k = 2$ requires $q^{-2} + 1 + O(q)$, because the stress tensor contributes at $h = 2$. Since the constant term of $\mathcal{J}^2$ is $2(196884) = 393768$, we obtain

$$\begin{aligned} Z_1 &= \mathcal{J}, \\ Z_2 &= \mathcal{J}^2 - 393767 = q^{-2} + 1 + 42987520q + 40491909396q^2 + \cdots, \\ Z_3 &= \mathcal{J}^3 - 590651\mathcal{J} - 64481279 \\ &= q^{-3} + q^{-1} + 1 + 2593096794q + \cdots. \end{aligned} \tag{17}$$

[1, §3.1] [checked by computation]At $k = 2$ the $h = 3$ vacuum descendant $L_{-3}|0\rangle$ accounts for one of the 42987520 states; the candidate number of new primaries there is 42987519. Counting all states and counting primaries are distinct even in this small example. [checked]

The leading large positive-energy growth of these functions follows from their vacuum pole under $S : \tau \mapsto -1/\tau$. With a chiral inverse temperature $b > 0$, $q = e^{-b}$, modular invariance gives $Z_k(b) \sim \exp(4\pi^2 k/b)$ at high temperature. Extracting the coefficient at chiral cylinder energy n uses a saddle of $bn + 4\pi^2 k/b$, at $b_* = 2\pi\sqrt{k/n}$, and gives

$$\log d_k(n) \sim 4\pi\sqrt{kn}. \tag{18}$$

For two chiralities this reproduces $2\pi\sqrt{cE_L/6} + 2\pi\sqrt{cE_R/6}$, the BTZ entropy. This is an asymptotic statement, controlled for large n at fixed k; a uniform large-k statement requires separate estimates. [1, §3.1][2, §2] [checked]

Exercise 1. Construct Z_2 and Z_3 by matching the vacuum character through q^0. Then compute $\mathcal{J}^2 - Z_2$ and explain why the tensor product of two Monster theories is a holomorphic $c = 48$ CFT but is not an extremal one. The discrepancy 393767 is precisely the excess number of weight-two states in that tensor product. [checked]

1.5 Existence constraints and nongeometric configurations

A torus trace only records graded dimensions. A CFT must also have associative operator products and consistent sewing on higher-genus surfaces. Sewing a genus-two surface, for example, probes sums of products of three-point coefficients. They contain information that no polynomial in $\mathcal{J}$ by itself supplies. Gaiotto–Yin constructed genus-two candidates at $k = 1, 2, 3$ and checked degeneration constraints; their results support consistency at that level and are not a general no-go theorem. [13]

Higher consistency conditions can therefore obstruct a candidate that passes the torus test, although the known obstructions have specific hypotheses. An extremal $c = 48$ CFT cannot possess Monster symmetry. An earlier proposed large-k nonexistence argument based on modular differential equations would have ruled out $k \geq 42$ under an additional assumption about modular differential equations, but the inference needed for that assumption was subsequently disproved and cannot be used as a general theorem. There are separate modular obstructions to extremal $\mathcal{N} = (2, 2)$ theories, which concern a supersymmetric elliptic genus and should not be transferred to bosonic extremal CFTs. [15, 14, 16]

A later result of Lin–Pei gives a further obstruction conditional on the Segal–Stolz–Teichner relation between supersymmetric field theories and topological modular forms. It requires the constant coefficient of a holomorphic $c = 24k$ partition function to be divisible by $24/\gcd(24, 2k)$. The extremal coefficients at $k = 2, 3, 4, 5$ are respectively $1, 1, 2, 2$, which violate the required divisibilities $6, 4, 3, 12$. This excludes those candidates under that conjectural input; the condition does not exclude all larger k, and is vacuous when $12 \mid k$. [17] [checked]There is thus neither a general construction of the proposed extremal family in these references nor a blanket nonexistence result for chiral CFTs above $c = 24$. Ordinary nonextremal examples, including tensor products, certainly exist. The existence issue here is the exceptionally large vacuum gap.

There is also a bulk obstruction to identifying the gauge and metric path integrals. The relation $e = \ell(A - \bar{A})/2$ allows $e = 0$ when $A = \bar{A}$, even though both are perfectly valid flat connections. In the complexified theory, arbitrary independent connections need not give a real invertible Lorentzian dreibein after continuation. [1, §§1.1,2.4] The topology problem is visible without solving those connections. If γ labels a solid-torus filling, a metric sum pairs both chiralities on that same filling, whereas a factorized prescription would require

$$\sum_{\gamma} F_\gamma(\tau)\overline{F_\gamma(\tau)} \quad \longrightarrow \quad \sum_{\gamma,\gamma'} F_\gamma(\tau)\overline{F_{\gamma'}(\tau)}. \tag{19}$$

The extra off-diagonal terms have incompatible choices of contractible cycle for the two connections. Their formal classical action is

$$I_{\gamma,\gamma'} = 2\pi i k\big(\gamma\tau - \gamma'\bar{\tau}\big), \tag{20}$$

which is generically complex even for a real Euclidean boundary torus. They do not correspond to ordinary real smooth fillings; nor are they known to obey the reality and regularity conditions needed for a physical Lorentzian spacetime. A possible completion could include such terms, but

a family of complex Einstein metrics realizing all of them has not been established. [7, §4.2] Complex saddles can enter a well-defined contour prescription, so the mere word "complex" does not exclude a saddle. What remains unestablished is a gravitational contour and interpretation that justify this enlarged sum. The factorized ansatz therefore remains a proposed quantum completion; it has not been derived by quantizing real nondegenerate metrics alone. [1, §2.4][7, §4.2]

2 Protected indices and the corrected Farey-tail expansion

2.1 The elliptic genus and its polar data

A microscopic partition function is naturally expanded as a sum over states of definite energy and charge. Its gravitational description instead suggests an expansion about classical geometries, each weighted by its action and dressed by quantum excitations. To compare these descriptions, we need to reorganize the state sum into a form in which the semiclassical contributions are visible. The Rademacher expansion does this for the protected indices considered here: it reconstructs the partition function from modular images of its low-energy, polar part. The images have a hierarchy of exponential weights in a black-hole regime, so the exact reconstruction also gives an expansion suited to a semiclassical interpretation. We will write this sum explicitly and identify the bulk contribution represented by each factor. [4, §4.1][2, §§2,5][3]

The D1–D5 system provides a first example with both a microscopic CFT and a protected observable. Its index lets us test this reorganization before attempting the unprotected partition function of pure gravity. [2]

For the compact $\mathcal{N}=(4,4)$ sigma model on $\mathrm{Sym}^m(K3)$, with $c_L=c_R=6m$, choose an $\mathcal{N}=2$ subalgebra and normalize its left charge $Q_L=2J_0^3$. Define

$$\phi_m(\tau,z)=\mathrm{Tr}_{RR}(-1)^{F_L+F_R}q^{L_0-c_L/24}\bar{q}^{\bar{L}_0-c_R/24}u^{Q_L},\qquad u=e^{2\pi iz}. \tag{21}$$

Both sectors have Ramond boundary conditions. Right supercharges pair states with opposite fermion parity at positive $\bar{L}_0-c_R/24$, so they cancel in the trace. For a compact theory with a discrete spectrum, only right Ramond ground states remain, and the result is holomorphic in τ. The left charge insertion commutes with this cancellation. Its coefficients are signed BPS indices, rather than the total number of states in the Hilbert space. [2, §3][3, §2.1]

The function is a weak Jacobi form of weight zero and index m:

$$\begin{aligned}\phi_m\left(\frac{a\tau+b}{r\tau+d},\frac{z}{r\tau+d}\right)&=\exp\left(\frac{2\pi imrz^2}{r\tau+d}\right)\phi_m(\tau,z),\qquad ad-br=1,\\ \phi_m(\tau,z+\lambda\tau+\mu)&=e^{-2\pi im(\lambda^2\tau+2\lambda z)}\phi_m(\tau,z),\qquad \lambda,\mu\in\mathbb{Z}.\end{aligned} \tag{22}$$

"Weak" means that its expansion $\sum_{n\geq 0,l\in\mathbb{Z}}c_m(n,l)q^nu^l$ has no negative n in the Ramond convention. The second equation is the elliptic transformation, reflecting spectral flow. It relates

$$(n,l)\longmapsto(n+\lambda l+m\lambda^2,l+2m\lambda),\qquad D\equiv 4mn-l^2\quad\text{unchanged}. \tag{23}$$

Hence the independent coefficients are labelled by D and l modulo $2m$. Terms with $D<0$ are called polar. Although $n\geq 0$, their charge-subtracted excitation energy $n-l^2/(4m)$ is negative. [2, §§3–4][3, §2.1] [checked]

This subtraction has a bulk meaning. The boundary current algebra accounts for a fixed energy carried by the charge. A charged BTZ horizon requires nonnegative excitation energy

after that current contribution is removed. In the BPS ensemble, polar states are below this charged black-hole threshold and seed the modular reconstruction of heavier states. The leading charged Cardy entropy is

$$S \sim 2\pi\sqrt{m\left(n-\frac{l^2}{4m}\right)} = \pi\sqrt{D}. \tag{24}$$

It applies in an appropriate large-charge asymptotic regime and is an entropy interpretation of an index only when cancellations do not erase the leading growth. [2, §§5–6]

2.2 The Rademacher expansion as a sum over saddle contributions

Begin by suppressing charges to isolate how a vacuum contribution becomes a sum over geometries. For this preliminary example, take a chiral counting function $F(\tau)$ that is holomorphic in the upper half-plane, may have a pole at the cusp, and has modular weight w with trivial scalar multiplier. Its integer-power expansion takes the form

$$F(\tau) = \sum_{N\geq 0} d(N) q^{N-c_L/24}, \qquad F^-(\tau) = \sum_{0\leq N<c_L/24} d(N) q^{N-c_L/24}. \tag{25}$$

The superscript minus retains the negative cylinder energies. These are precisely the terms that diverge at $q = 0$, the low-temperature cusp. Here N is a nonnegative integer and $c_L/24$ is an integer, as required by the assumed invariance under T. The charged example below permits fractional exponents through a nontrivial multiplier. In the gravity interpretation, $N = 0$ is the AdS vacuum and $0 < N < c_L/24$ describes excitations that remain below the chiral black-hole threshold. The coefficients are indexed multiplicities when F belongs to a protected trace. [4, eqs. (4.1)–(4.4) and §5.1]

The modular transformation law $F(\gamma\tau) = (r\tau + d)^w F(\tau)$ specifies the polar behavior near every modular image of that cusp. It suggests reconstructing F by putting the known polar terms at each image:

$$F(\tau) \overset{\text{formal}}{=} \sum_{\gamma\in\Gamma_\infty\backslash \mathrm{PSL}(2,\mathbb{Z})} (r\tau + d)^{-w} F^-(\gamma\tau), \qquad \gamma = \begin{pmatrix} a & b \\ r & d \end{pmatrix}, \quad ad - br = 1. \tag{26}$$

Here $\gamma\tau = (a\tau + b)/(r\tau + d)$ and $\Gamma_\infty = \langle T\rangle$, with $T : \tau \mapsto \tau + 1$, removes repetitions related by integer shifts of the transformed modulus. This formula displays the organization of the sum; convergence, constant terms, and possible holomorphic additions must still be treated. For the negative-weight coefficient functions relevant below, compatible polar data determine the answer uniquely after the corrected regularization is supplied. [4, §4.1][3, §§3–4]

The exponent already explains the semiclassical interpretation. A term in the transformed polar part is

$$d(N)e^{2\pi i(N-c_L/24)\gamma\tau} = e^{-I_\gamma^{\text{chiral}}}\, d(N)e^{2\pi i N\gamma\tau}, \qquad I_\gamma^{\text{chiral}} = \frac{2\pi i c_L}{24}\gamma\tau. \tag{27}$$

The vacuum energy fixes the leading action, while the remaining factor counts excitations propagating in that background. This is a chiral weight in the protected ensemble; the ordinary Einstein thermal partition function in lecture 6 includes both chiralities. For the identity image it is the thermal-AdS vacuum weight. For the S image, $\gamma\tau = -1/\tau$, it is the black-hole weight. Other images correspond to other choices of contractible boundary cycle. The same light spectrum is

evaluated in each filling, rather than assigning a separate geometry to each microscopic state. [4, §5.1][2, §§5–6]

For example, the S image of the vacuum at $\tau = iy$ is $\exp[2\pi c_L/(24y)]$. Extracting a large positive-energy coefficient from this term produces Cardy growth. The other modular denominators give smaller exponentials in the corresponding Rademacher coefficient expansion, as the example below will show. Primitive pairs (r, d) can be organized by rational fractions with increasing denominator; the resulting subleading contributions form the "Farey tail." Thus the usefulness of the expansion is both microscopic and semiclassical: it determines heavy coefficients from light ones while displaying their hierarchy of saddle contributions. [4, §4.1][2, §2] [checked]

Restoring charges: the full partition-function summand

The chemical potentials in the elliptic genus specify boundary gauge holonomies. Their transformation cannot be recovered from the metric action alone. Spectral flow separates the indexed multiplicities from the charge-lattice sum:

$$\phi_m(\tau, z) = \sum_{\mu \pmod{2m}} h_\mu(\tau)\theta_{m,\mu}(\tau, z), \quad \theta_{m,\mu}(\tau, z) = \sum_{l \equiv \mu \ (2m)} q^{l^2/(4m)} u^l. \tag{28}$$

Each theta function sums charges in one spectral-flow class. Write

$$h_\mu(\tau) = \sum_\nu b_\mu(\nu) q^\nu, \qquad h_\mu^-(\tau) = \sum_{\nu<0} b_\mu(\nu) q^\nu, \qquad \nu = n - \frac{l^2}{4m}. \tag{29}$$

The coefficient $b_\mu(\nu)$ is the indexed multiplicity after the charge energy has been removed; fractional ν are allowed. The polar seed retains the negative ν terms and all their charge-lattice copies. In the NS description these include the vacuum and light BPS excitations. The seed is consequently larger than the pure-gravity vacuum character. [2, §3][3, §2.1]

The general charged formula is most transparent before specializing the weight. Let $Z(\tau, \boldsymbol{z})$ be a holomorphic Jacobi partition function of weight W, with theta decomposition $Z = \sum_\mu h_\mu \Theta_\mu$ on a positive-definite charge lattice of rank N_Q. The theta functions have weight $N_Q/2$, so the coefficient vector has weight $w = W - N_Q/2$. Thus even a weight-zero index can require reconstruction of negative-weight coefficients. [4, §4.1] Define its charge quadratic form $Q(\boldsymbol{z})$ by the transformation law

$$Z\left(\gamma\tau, \frac{\boldsymbol{z}}{r\tau + d}\right) = (r\tau + d)^W \exp\left(\frac{\pi i r Q(\boldsymbol{z})}{r\tau + d}\right) Z(\tau, \boldsymbol{z}). \tag{30}$$

For our one-charge elliptic genus, $W = 0$, $\boldsymbol{z} = z$, and $Q(z) = 2mz^2$. With these conventions the full modular-image expansion is

$$Z(\tau, \boldsymbol{z}) \overset{\text{formal}}{=} \sum_{\gamma \in \Gamma_\infty \backslash \mathrm{PSL}(2,\mathbb{Z})} \frac{\exp\left[-\dfrac{\pi i r Q(\boldsymbol{z})}{r\tau + d}\right]}{(r\tau + d)^W} \sum_\mu h_\mu^-\left(\frac{a\tau + b}{r\tau + d}\right) \Theta_\mu\left(\frac{a\tau + b}{r\tau + d}, \frac{\boldsymbol{z}}{r\tau + d}\right). \tag{31}$$

This is the partition-function expansion of [4, eq. (4.10)], with its chemical potentials denoted by $\boldsymbol{z}$ and its lower-left matrix entry by r. We write the left coset because it is the one that identifies representatives with the same contractible cycle in our convention. The word "formal" records the need to regulate the negative-weight coefficient sum; the explicit corrected equality

is given below. The general formula here assumes the scalar Jacobi transformation law (30); any additional scalar multiplier must also be included in the modular prefactor. [4, §4.1][3, §§3,5]

Every factor has a role in the bulk interpretation. The transformed modulus specifies the filling. The polar character includes its vacuum weight and the gas of light excitations around it. The theta function sums its charge or gauge-flux sectors, evaluated at the transformed boundary holonomies. The exponential outside the seed compensates the Jacobi phase and represents the corresponding quadratic gauge-field contribution. Finally, $(r\tau+d)^{-W}$ supplies the modular weight of the full observable. These factors must transform together: keeping the vacuum exponential while omitting the theta function or its compensating phase would compute a different object. [4, §§4.1,5.1][3, §5.2]

In particular, the D1–D5 elliptic genus has the concrete expansion

$$\phi_m(\tau,z) \overset{\text{formal}}{=} \sum_{\gamma\in\Gamma_\infty\backslash \mathrm{PSL}(2,\mathbb{Z})} e^{-2\pi i m r z^2/(r\tau+d)} \sum_{\mu \pmod{2m}} h_\mu^-(\gamma\tau)\theta_{m,\mu}\left(\gamma\tau, \frac{z}{r\tau+d}\right). \tag{32}$$

Writing $q_\gamma = e^{2\pi i\gamma\tau}$, an individual polar frequency and charge in this sum contribute

$$b_\mu(\nu)\, e^{-2\pi i m r z^2/(r\tau+d)} q_\gamma^{\nu+l^2/(4m)} e^{2\pi i l z/(r\tau+d)}, \qquad \nu<0, \quad l\equiv\mu \pmod{2m}. \tag{33}$$

Thus the charge-subtracted excitation energy ν, the charge energy $l^2/(4m)$, and the chemical-potential insertion all propagate with the same transformed cycles. This gives an explicit saddle summand for the string-theory index, including its light-state and gauge contributions. [4, eq. (4.10) and §5.1][2] [checked]

We can also derive why the charge-sector mixing matrices have disappeared from (32). The vector h transforms with weight $w=-1/2$ and a multiplier matrix ρ:

$$h(\gamma\tau) = (r\tau+d)^w \rho(\gamma) h(\tau), \qquad w=-\tfrac{1}{2}. \tag{34}$$

The theta vector transforms in the dual representation,

$$\boldsymbol{\theta}_m\left(\gamma\tau, \frac{z}{r\tau+d}\right) = (r\tau+d)^{1/2} e^{2\pi i m r z^2/(r\tau+d)} \rho(\gamma)^{-T} \boldsymbol{\theta}_m(\tau,z). \tag{35}$$

Here $-T$ means inverse transpose. Half-integer powers and the multiplier matrices are defined with the same branch convention, $-\pi < \arg(r\tau+d) \le \pi$, so their phases are consistent. [3, eqs. (2.12)–(2.14)] Substituting this law into a summand of (32) gives

$$e^{-2\pi i m r z^2/(r\tau+d)} \sum_\mu h_\mu^-(\gamma\tau)\theta_{m,\mu}\left(\gamma\tau, \frac{z}{r\tau+d}\right) = (r\tau+d)^{-w} \boldsymbol{\theta}_m(\tau,z)^T \rho(\gamma)^{-1} h^-(\gamma\tau). \tag{36}$$

The right side is exactly the modular image of the vector of polar characters, contracted with the original theta functions. This explains the passage from the vector-valued Rademacher formula to the full partition function in (31). It also identifies where to insert the regulator without changing the indexed degeneracies. [4, §4.1][3, §2.1] [checked]

The correction and the explicit regulated sum

The obstacle to interpreting the displayed sums as exact equalities is their convergence. The polar-character weights are negative in both string-theory examples. The original construction

attempted to raise those weights first, using a derivative identity: at integral $w \le 0$, $D_\tau^{1-w} f$ transforms with weight $2-w$ when f has weight w, where $D_\tau = (2\pi i)^{-1}\partial_\tau$. At precisely this integer order the additional chain-rule terms cancel. This is Bol's identity, and it suggests passing to a positive weight where ordinary Poincaré sums have better convergence. [4, eq. (4.11)][3, §5.1]

For a charged Fourier series, the operator measuring its charge-subtracted frequency is

$$\mathcal{D}_m = D_\tau - \frac{1}{4m} D_u^2, \qquad D_u = u\partial_u = (2\pi i)^{-1}\partial_z, \qquad \mathcal{D}_m(q^n u^l) = \left(n - \frac{l^2}{4m}\right) q^n u^l. \tag{37}$$

It is called a heat operator because it combines one derivative in τ with two in z; it annihilates the theta functions. For the weight-$-1/2$ vector h, the proposed weight-raising operation has order $3/2$. Multiplying coefficients by $|\nu|^{3/2}$ defines a formal transform, but the integer-order identity does not establish its modular covariance. The transformed Jacobi form does not have the claimed transformation law, and the attempted full Poincaré extension of the original truncated expression gives zero. The coefficient reconstruction and its semiclassical interpretation survive; the fractional-transform justification does not. [3, §5.1][2, version 3, note added]

The corrected construction instead regulates each original polar frequency. For $\nu < 0$ and $r \neq 0$, define

$$R_w(X) = \frac{1}{\Gamma(1-w)} \int_0^X e^{-v} v^{-w} \mathrm{d}v, \qquad X_{\gamma,\nu} = \frac{2\pi i|\nu|}{r(r\tau + d)}. \tag{38}$$

The nonidentity vector summand becomes

$$(r\tau + d)^{-w} \rho(\gamma)^{-1} b(\nu) e^{2\pi i \nu \gamma\tau} R_w(X_{\gamma,\nu}). \tag{39}$$

Here $b(\nu)$ is the vector of coefficients in (29), with zero entries in sectors where that frequency is absent. Define $\mathcal{C}_K$ to contain the identity and the representatives with $0 < r \le K$, $|d| \le K$, $(r,d) = 1$. The cutoff is symmetric in d. The corrected full elliptic genus is

$$\begin{aligned}\phi_m(\tau, z) = {} & \frac{1}{2} \sum_\mu b_\mu(0) \theta_{m,\mu}(\tau, z) \\ & + \lim_{K\to\infty} \sum_{\gamma \in \mathcal{C}_K} e^{-2\pi i m r z^2/(r\tau+d)} \sum_\mu \theta_{m,\mu}\left(\gamma\tau, \frac{z}{r\tau+d}\right) \sum_{\nu<0} b_\mu(\nu) e^{2\pi i \nu\gamma\tau} R_{-1/2}(X_{\gamma,\nu}).\end{aligned} \tag{40}$$

For the identity term, the regulator is defined to be one. The sums on μ run over the charge classes modulo $2m$. The coefficient $b_\mu(0)$ is the constant term of h_μ; it vanishes when zero frequency is forbidden in that sector. These constant coefficients are determined by the polar data through the Rademacher expansion, not adjustable additions. The explicit half-constant term is part of the summation prescription. Using representatives in PSL counts γ and $-\gamma$ together; equivalently one may use the full SL coset sum with a factor of $1/2$, as in the source. Equation (36), applied to each regulated polar frequency, converts (40) into the corrected vector formula. [3, eqs. (3.4)–(3.7) and Appendix A] [checked]

The regulation is compatible with the leading saddle interpretation. For distant images, $X_{\gamma,\nu}$ is small and $R_w(X) \sim X^{1-w}/\Gamma(2-w)$ suppresses the divergent tail. At $w = 0$ it subtracts the limiting cusp value:

$$e^{2\pi i\nu\gamma\tau} R_0(X_{\gamma,\nu}) = e^{2\pi i\nu\gamma\tau} - e^{2\pi i\nu a/r}. \tag{41}$$

In a semiclassical saddle regime with large positive $\operatorname{Re} X$, $R_w(X)$ approaches one exponentially fast. The vacuum action, charge-lattice factor, and light excitations in (31) therefore remain

the leading constituents of each saddle contribution. The correction supplies a valid sum for the original index instead of replacing its degeneracies by fractionally differentiated coefficients. [3, §§3,5] [checked]

Finally, a proposed polar spectrum must satisfy modular consistency. Arbitrary polar coefficients can produce an anomaly under modular transformations; cusp forms of dual weight $2-w$ and conjugate multiplier impose linear constraints on them. A cusp form is a holomorphic modular form that vanishes at the cusp. An actual elliptic genus satisfies these constraints. At negative weight, compatible polar data fix the holomorphic result uniquely, since there is no nonzero holomorphic negative-weight form without a cusp pole to add. This is the precise setting in which the reconstruction from the light spectrum is sufficient. [3, §4]

A direct example of the coefficient reconstruction is the Rademacher expansion for $\mathcal{J}(\tau) = q^{-1} + \sum_{n>0} a(n)q^n$:

$$a(n) = \frac{2\pi}{\sqrt{n}} \sum_{r=1}^{\infty} \frac{K(n,-1;r)}{r} I_1\left(\frac{4\pi\sqrt{n}}{r}\right), \qquad K(n,\nu;r) = \sum_{\substack{d \pmod r \\ (d,r)=1}} e^{2\pi i(nd+\nu d^{-1})/r}. \tag{42}$$

Here d^{-1} is the multiplicative inverse modulo r and I_1 is the modified Bessel function. These exponential sums are called Kloosterman sums. The $r=1$ term gives $e^{4\pi\sqrt{n}}$, since $I_1(X) \sim e^X/\sqrt{2\pi X}$; larger r give successively smaller exponentials. The exact convergent coefficient formula refines the asymptotic Cardy estimate. Individual summands need not be integers; the complete sum is the integer CFT coefficient. The vector-valued analogue uses I_{1-w} and the multiplier matrices. [2, §2][3, Appendix A]

2.3 Why the modular images describe different bulk fillings

Choose a spatial cycle A and a Euclidean-time cycle B on the boundary torus, with periods 2π and $2\pi\tau$. A smooth solid torus fills in one primitive cycle

$$C_{r,d} = dA + rB, \qquad (r,d) = 1, \tag{43}$$

making it contractible. An auxiliary cycle $bA+aB$ with $ad-br=1$ completes a basis; the modulus in that basis is $\gamma\tau = (a\tau+b)/(r\tau+d)$. Changing the auxiliary cycle by an integer multiple of the contractible one does not change the filling. Thus the fillings are indexed by $\Gamma_\infty\backslash\mathrm{PSL}(2,\mathbb{Z})$, where $\Gamma_\infty = \langle T\rangle$ and $T:\tau\mapsto\tau+1$. [2, §5][7, §2.1]

For $(r,d)=(0,1)$, the spatial circle contracts and the saddle is thermal AdS_3. For $(r,d)=(1,0)$, the thermal circle contracts and the saddle is the Euclidean BTZ geometry. The other primitive cycles are modular images of these. In the supersymmetric problem the choice of spin structure and R-symmetry gauge holonomy must also extend smoothly. The gauge fields and spectral-flow sectors encoded in (28) are therefore part of the saddle data; the elliptic genus is not obtained from the metric action alone. [2, §§5–6][3, §5.2]

The summand of (40) now has a geometric label: it belongs to the filling whose meridian is $C_{r,d}$. Its polar exponential propagates the light spectrum around the noncontractible cycle, while its transformed theta function and compensating Jacobi phase supply the gauge contribution. The identity image retains the light-state expansion of thermal AdS; its S image gives the BTZ contribution. Summing the images reconstructs the heavy BPS coefficients of the same microscopic index. At large charges the dominant exponential selects a saddle, and its changes across modular images give a Farey pattern of phase boundaries. The corrected regulator preserves this

leading semiclassical picture, while altering the precise subleading expression and enforcing the polar consistency constraints. A term-by-term derivation of its entire measure from a fundamental gravity contour is not established by these papers. [2, §§5–6][3, §§5.2–5.3]

2.4 The relation to de Boer's supergravity spectrum

The polar seed is not just an abstract list of modular coefficients. De Boer studied the Kaluza–Klein spectrum of supergravity on $\mathrm{AdS}_3 \times S^3$ and its organization into short representations of $SU(1,1|2)_L \times SU(1,1|2)_R$. The multiparticle chiral-primary spectrum agrees with that of the symmetric-product K3 theory at large m. [5] It also explains how some states that appear as additional CFT operators are multiparticle supergravity states, rather than new elementary bulk fields.

The finite-m elliptic genus needs more care. An unrestricted supergravity Fock space overcounts states, and taking $m \to \infty$ at the start makes the relevant elliptic-genus comparison divergent. In [6], each short multiplet and its descendants are assigned a degree d_i, and only multiparticle states with

$$\sum_i d_i \leq m \tag{44}$$

are retained. This is a proposed implementation of the stringy exclusion principle, fixed first by the finite-m chiral-primary spectrum. Its extension to all descendants is an assumption subsequently tested by the index comparison, rather than a deduction from linearized supergravity alone. [6, §2]

With this prescription, de Boer finds exact agreement of the elliptic genera in the NS left-moving energy range

$$h_{\mathrm{NS}} \leq \frac{m+1}{4}. \tag{45}$$

This protected agreement tests the light-state index rather than the full thermal spectrum. To relate the tested range to polar data, spectral flow by half a unit gives $h_{\mathrm{NS}} = n - l/2 + m/2$ in one choice of flow direction, with n, l the Ramond variables of (21). The invariant discriminant $4mn - l^2$ then identifies the low-energy sectors that enter the Farey-tail seed. A cutoff on h_{NS} and a condition on that discriminant should not be silently identified at all charges. [6, §3][2, §3] [checked]

The microscopic index can be computed explicitly using the symmetric-product formula. If $\phi_{K3} = \sum_{n,l} c_{K3}(n,l) q^n u^l$, then

$$\sum_{m \geq 0} p^m \phi_{\mathrm{Sym}^m K3}(\tau, z) = \prod_{s \geq 1} \prod_{n \geq 0} \prod_{l \in \mathbb{Z}} (1 - p^s q^n u^l)^{-c_{K3}(sn,l)}. \tag{46}$$

Here p counts the symmetric-product degree, and s is a cycle length. The factorized generating function has the structure of a second-quantized index. De Boer's exclusion rule cuts the bulk multiparticle space to the same fixed degree. The Farey-tail reconstruction then uses suitable low-energy indexed data and modularity to obtain the heavy indexed coefficients. These are successive steps in the same comparison: identify the light bulk states, test their protected counting, and reorganize the full index as modular images. [18][6, §§2–5][2]

2.5 Attractor black holes and the scope of the motivation

The D1–D5 example could leave the impression that the geometric reconstruction depends on the special holomorphic K3 elliptic genus. A useful second example tests it in a different compactifi-

cation, with an indefinite charge lattice and a partly nonholomorphic index. The same sequence of steps remains available: identify light indexed states, separate their charge dependence, and reconstruct their modular images. This gives additional string-theory evidence for the saddle organization that we will try to apply to pure gravity. [4, 3]

Consider an M5-brane wrapped on a smooth ample divisor of a compact Calabi–Yau threefold X of $SU(3)$ holonomy. Write $P = p^a\alpha_a$ for its Poincaré-dual cohomology class. Ample means that this integral class can be represented by a Kähler form; this positivity assumption is part of the compactification used in the reconstruction. The effective string has an $\mathrm{AdS}_3 \times S^2$ near-horizon region and a $(0,4)$ CFT; compactifying its spatial circle relates this system to four-dimensional D4–D2–D0 attractor black holes. Here "attractor" refers to the fixing of the vector-multiplet moduli at the horizon by the black-hole charges. In this compactification the Kähler form is aligned with P; the claim is not that every scalar modulus is fixed. [4, §§2–3][3, §2.2] If $d_{abc} = \int_X \alpha_a \wedge \alpha_b \wedge \alpha_c$ and $D = \frac{1}{6} d_{abc} p^a p^b p^c$, the left and right central charges are

$$c_L = 6D + c_2(X)\cdot P, \qquad c_R = 6D + \tfrac{1}{2} c_2(X)\cdot P. \tag{47}$$

The terms involving the second Chern class $c_2(X)$ are the anomaly contributions. [4, §§2–3]

With the paper's charge convention $d_{ab} = -d_{abc}p^c$, large gauge transformations act as

$$q_a \mapsto q_a + d_{ab}s^b, \qquad q_0 \mapsto q_0 + s^a q_a + \tfrac{1}{2} d_{ab} s^a s^b, \qquad \widehat{q}_0 = q_0 - \tfrac{1}{2} d^{ab} q_a q_b \quad \text{invariant.} \tag{48}$$

Here q_a are M2 charges, q_0 is the momentum, $s^a \in \mathbb{Z}$ labels the large gauge transformation, and d^{ab} is the inverse form. Expanding the last expression after the shift directly checks its invariance. At leading order the entropy is $S = 2\pi\sqrt{D\widehat{q}_0}$ in these conventions, so $\widehat{q}_0 < 0$ supplies the polar data. [4, eqs. (2.8)–(2.14)] [checked]

The graded trace with $(-1)^F$ but without F^2 vanishes: the four fermion zero modes pair states whose contributions have opposite signs. A nonzero protected observable is instead a modified elliptic genus with an $F^2(-1)^F$ insertion, where $F = 2\bar{J}_0^3$; the charge phase $e^{\pi i p^a q_a}$ required in that convention must also be retained. Explicitly its charged trace is

$$Z'_p = \mathrm{Tr}_{\mathrm{R}}\left[F^2(-1)^F e^{\pi i p^a q_a} q^{L_0 - c_L/24} \bar{q}^{\bar{L}_0 - c_R/24} e^{2\pi i z^a q_a}\right]. \tag{49}$$

The trace uses Ramond boundary conditions for the right-moving fermions. The insertion saturates the fermion zero modes that would otherwise make the index vanish.[2] To expose the modular reconstruction, remove the universal noncompact center-of-mass factor and denote the remaining index by Z_p. Its charge dependence separates as

$$Z_p(\tau,\bar{\tau},z) = \sum_\mu H_\mu(\tau)\Theta_{p,\mu}(\tau,\bar{\tau},z), \qquad H_\mu(\tau) = \sum_\nu d_\mu(\nu) q^\nu. \tag{50}$$

Here μ labels the finitely many charge classes under the large gauge transformations in (48). The theta function $\Theta_{p,\mu}$ sums the charges in one such class, with their left- and right-moving charge energies, chemical potentials z^a, and the charge phase already present in Z'_p. The allowed charges lie in a shifted dual lattice: in lower-index notation, $q_a \in \Lambda^* + \frac{1}{2} d_{ab} p^b$. Here Λ is the integral spectral-flow lattice with quadratic form d_{ab}, and Λ^* is its dual charge lattice. The shift is included in the theta function, together with its phase. [4, eqs. (3.21), (3.29)–(3.33)] The lattice

[2] We use the F^2 normalization of [4, eq. (3.14)]. The convention in [3, eq. (2.15)] instead inserts $F^2/2$. This overall factor of two is absorbed in H_μ and d_μ when using the latter paper's reconstruction formula; it changes neither the modular weight nor the regulator.

has indefinite signature, so the convergent sum uses the positive left- and right-moving quadratic forms separately, with factors of both q and $\bar{q}$. Such a sum is called a Siegel–Narain theta function. The functions H_μ count the remaining excitations: ν is the spectral-flow-invariant momentum $\widehat{q}_0$ in the above convention, including the vacuum energy shift, and $d_\mu(\nu)$ is its indexed multiplicity. [4, §3.2 and Appendix A][3, §2.2]

The reconstruction in these references assumes that H is holomorphic in the upper half-plane and transforms as a vector-valued modular form, with its only poles at the cusp. Under those assumptions the part to reconstruct is H; the Siegel–Narain factor retains the nonholomorphic charge dependence. In the conventions with the center-of-mass factor removed, the weight of H is $w_p = -1 - b_2(X)/2$, where $b_2(X)$ is the number of independent harmonic two-forms. Let ρ_p be the modular mixing matrix of its charge classes. The counterpart of (40), written with the charge-mixing matrices explicit, is

$$\begin{aligned} Z_p(\tau,\bar{\tau},z) = \frac{1}{2}\sum_{\mu} d_\mu(0)\Theta_{p,\mu}(\tau,\bar{\tau},z) \\ + \lim_{K\to\infty}\sum_{\gamma\in\mathcal{C}_K}(r\tau+d)^{-w_p}\sum_{\mu,\sigma}\Theta_{p,\mu}(\tau,\bar{\tau},z)[\rho_p(\gamma)^{-1}]_{\mu\sigma} \\ \times\sum_{\nu<0} d_\sigma(\nu)e^{2\pi i\nu\gamma\tau}R_{w_p}(X_{\gamma,\nu}). \end{aligned} \tag{51}$$

The identity prescription and the half-constant term are the same as in (40). As in that example, the polar data must satisfy the modular consistency constraints. If an index instead requires a nonholomorphic completion of its coefficient functions, this holomorphic formula alone does not reconstruct it; such anomalies are discussed in [3, §6]. The powers $e^{2\pi i\nu\gamma\tau}$ again display the light-state contribution evaluated in each filling. Transforming the Siegel–Narain theta functions rewrites the same summand with theta functions at the transformed modulus and gauge holonomies, as in (31); their antiholomorphic weight and right-moving chemical potentials must now be transformed as well. Keeping the mixing matrix explicit in (51) avoids treating this indefinite-lattice index as an ordinary holomorphic Jacobi form. [3, §2.2 and eq. (3.4)][4, §§3.2,4.1 and erratum]

The terms admit an interpretation as Euclidean BTZ fillings decorated with gauge fluxes and light BPS excitations. In this compactification, wrapped M2-branes and anti-M2-branes contribute to the polar sector. Their indexed counting supplies the seed of (51); it is not an extra factor that can be omitted while keeping the same observable. The corrected regularization preserves this semiclassical interpretation, as the attractor paper's erratum also emphasizes. The two string-theory examples therefore support a sum over modular images with specified light states and gauge contributions. In both, the microscopic protected observable is known before its gravitational reorganization. [4, §5 and erratum][3, §5]

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